

Understanding the Lunar Reconnaissance Orbiter Camera

A Simple, Comprehensive Guide to the LROC Mission, Cameras, Data Products, and Processing

1. Introduction to the LRO Mission

NASA launched the Lunar Reconnaissance Orbiter (LRO) in June 2009 to create the most detailed map of the Moon ever made. The spacecraft travels around the Moon in a polar orbit, allowing its instruments to observe the entire lunar surface over time as the Moon rotates beneath it.

The mission has two main goals:

- **Exploration:** Finding safe landing sites for future human and robotic missions, searching for resources like water ice, and mapping surface hazards.
- **Science:** Studying lunar geology, measuring impact cratering rates over time, and understanding how the Moon's environment changes.

2. The LROC Camera Systems

To capture both extreme surface detail and wide global coverage, the Lunar Reconnaissance Orbiter Camera (LROC) suite uses three distinct camera units mounted together on the spacecraft:

Narrow Angle Cameras (NAC-L and NAC-R)

The system features two identical high-resolution telescopes called **NAC-Left** and **NAC-Right**. They are aimed slightly apart to double the width of the ground area captured in a single pass.

- **High Resolution:** Captures details down to 0.5 meters (1.6 feet) per pixel from a 50 km altitude. This makes it possible to spot small boulders, crater rims, and even Apollo moon landers.
- **Pushbroom Scanning:** Rather than taking a standard square photograph, the NACs record single lines of pixels continuously as the spacecraft moves forward over the Moon's surface.

Wide Angle Camera (WAC)

The **Wide Angle Camera** is a single wide-angle instrument designed to observe large regions at once.

- **Wide Coverage:** A single WAC image covers a swath about 105 km wide, allowing it to build complete global maps of the Moon every 28 days.
- **Multispectral Capabilities:** Captures light across 7 color bands (2 ultraviolet and 5 visible light wavelengths). This helps scientists map variations in surface minerals, such as titanium and iron abundances.

Feature	Narrow Angle Cameras (NAC)	Wide Angle Camera (WAC)
Primary Function	High-detail local imaging & topography	Global mapping & mineral composition
Resolution (at 50 km)	0.5 meters per pixel	75 to 384 meters per pixel

Image Type	Black & White (Monochrome line-scan)	Color & UV (Multispectral frame)
Swath Width	5 km combined width	105 km width

3. Understanding LROC Data Levels

Data downlinked from LRO undergoes sequential processing stages before scientists use it for analysis:

1. Raw Data (EDR - Experiment Data Record):

Unprocessed image files containing raw sensor values (Digital Numbers) directly converted from telemetry packets.

2. Calibrated Data (CDR - Calibrated Data Record):

Corrected images where camera artifacts (like dark noise and lens shading) are removed, converting raw counts into actual surface brightness and reflectivity values.

3. Derived Maps & Models (RDR - Reduced Data Record):

Finished products like seamless global mosaics, mineral composition maps, and 3D Digital Elevation Models (DEMs) created by matching overlapping images.

4. How Scientists Process and Use LROC Images

Processing raw spacecraft data into usable maps involves a few basic steps:

1. **Position Tracking:** Using SPICE positioning kernels to calculate exactly where the spacecraft was and where the camera was pointed when the shot was taken.
2. **Lighting Normalization:** Adjusting the brightness levels of images so that scenes captured at different times of day with different shadow angles can blend seamlessly together.
3. **Map Projection:** Aligning image pixels to precise lunar latitude and longitude coordinates so they fit accurately on a spherical globe.

Key Scientific Applications

- **3D Surface Topography:** Overlapping NAC photos taken from slightly different orbital angles are combined to measure surface heights and slope steepness.
- **Shadowed Crater Exploration:** Scientists use special long exposures to look inside permanently shadowed craters at the lunar poles, revealing hidden ice deposits illuminated only by faint starlight.
- **Monitoring New Impacts:** By comparing older photos with new ones, scientists detect fresh impact craters formed by meteoroid strikes during the mission.